Thinking Through a Space with a Personal AI

Research interests and initial PhD proposal

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Start at the entrance

Imagine walking into a small exhibition space with an AI assistant on your glasses. At the entrance, I point to an introductory panel: "If we turn this towards the doorway, what will visitors see first?" The assistant shows two positions. I walk towards the projection wall and reconsider: "I want them to read the title before noticing the projection. Keep this route open." It compares views from where I stand and updates the digital model as we settle on a layout.

On the next visit, a projector has moved. I want to pick up yesterday's discussion: remember why we placed the panel there, check the changed room, and work out what to keep. This is where I would begin studying a personal AI that can think through a place with its user.

Why I want to build this

My starting point is architectural work, from field records and CAD/Rhino models to lighting and projection at the Damo Cave exhibition. Working on site put physical space and digital systems in the same workflow. I want an AI I can talk things through with while looking at the place itself. My longer-term ambition resembles Jarvis: an assistant that comes to know my work, offers its own judgment, and helps make things happen. Smart glasses could let this conversation continue while I keep looking at the place in front of me.

Why Synteraction

TOM offers a wearable-assistant platform that connects information about the wearer and environment to assistance [1]. CRAFT carries real-world encounters into a fiction project and maintains an evolving fiction context across sessions [2]. PersonaMail supports later, context-sensitive use of communication preferences [3]. This is the part of Heads-Up Computing I want to explore: an assistant that keeps up while someone is looking around and doing the work.

Building on speech-driven AR authoring such as ImaginateAR [4], I would study how an assistant follows a designer through a changing room and revisits the reasons behind a layout on a later visit.

Two questions I would investigate

1. As a person moves and talks, how can an AI connect words such as "here" and "this panel" to the intended place, and help compare what different layouts would mean there?

2. When the person returns, how can the assistant use the reasons behind earlier choices to revise the layout for a changed room?

Build around one room

I would first observe designers walking through a small exhibition space, then review recordings with them to understand which observations lead them to change a layout. The prototype would use a measured room model with a movable panel and a defined projection area, with an existing vision-language model interpreting the conversation. Camera-equipped smart glasses would provide first-person views, with the wearer's position aligned to the room model. Speech and pointing would connect the conversation to objects in the room. Brief visual outlines would show alternatives against the real space, so the person could keep looking around while discussing a change.

The assistant would call a small set of tools to move or rotate the digital panel, compare views and save a layout version. Each choice would retain the relevant object, the observation point and the person's stated reason: for example, "read the title before seeing the projection". On a return visit, it could revisit that reason against the new equipment positions. When a reference is unclear, a brief highlight would show which object the assistant means. The person could point again or change their mind in ordinary conversation, then see the resulting model update.

Test it while people work

The main comparison would use the same model, environmental inputs and editing tools. Both assistants would be able to retrieve the same conversation, saved layout versions, viewpoints and stated reasons. Both would support clarification, visual previews, corrections and undo. The experimental assistant would additionally link each layout choice to its observation point and stated reason, and let the person revisit those links when the room changes. This tests whether organizing the history around design decisions helps beyond simply having the conversation available.

Participants with spatial design experience would work through equivalent layout briefs, then return after an equipment change. I would measure mistaken object references and the effort needed to correct them, and examine recordings of how participants compare and revise alternatives. Geometric checks on the room model could assess route obstructions and title visibility from specified viewpoints; independent designers could assess how well each solution meets the brief. The briefs would identify requirements that still apply after the change. I would score whether participants preserve these and revise the affected choices, recording mistaken carry-over. Decision time and interruptions to looking around the room would show the cost of the interaction. Observation and pilot results would guide the study size and session spacing.

What I bring to the work

Alongside spatial work, I have built a Pygame teammate that intercepts projectiles and signals its actions through text and sound. My machine-learning coursework involved training and comparing classifiers; in an eight-person capstone, I implemented a direct-detection baseline and investigated model errors. My AI background lets me build and investigate a working prototype, while keeping the research focused on how people collaborate with it. Wearable and AR development would be new to me; I would learn it alongside the first observation study. Repeated site visits are a practical starting point for that personal AI.

References

[1] Nuwan Janaka et al. (2024). TOM: A Development Platform For Wearable Intelligent Assistants. UbiComp Companion 2024. https://doi.org/10.1145/3675094.3678382

[2] Runze Cai et al. (2026). CRAFT: Exploring Wearable Creative AI on Smart Glasses for Fiction Writing in Real-World Contexts. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 10(3), Article 81. https://doi.org/10.1145/3831952

[3] Rui Yao et al. (2026). PersonaMail: Learning and Adapting Personal Communication Preferences for Context-Aware Email Writing. IUI 2026. https://doi.org/10.1145/3742413.3789123

[4] Jaewook Lee et al. (2025). ImaginateAR: AI-Assisted In-Situ Authoring in Augmented Reality. UIST 2025. https://doi.org/10.1145/3746059.3747635